PHONOLOGICA 1980

Akten der Vierten Internationalen Phonologie-Tagung

Wien, 29. Juni—2. Juli 1980

herausgegeben von

Wolfgang U. DRESSLER — Oskar E. PFEIFFER —
John R. RENNISON

unter redaktioneller Mitarbeit von

G. Dogil

Innsbruck 1981

Die INNSBRUCKER BEITRÄGE ZUR SPRACHWISSENSCHAFT werden gefördert durch das Bundesministerium für Wissenschaft und Forschung in Wien, die Kulturabteilung des Amtes der Tiroler Landesregierung und durch den Universitätsbund Innsbruck.

CIP-Kurztitelaufnahme der Deutschen Bibliothek
Phonologica: Akten d. ... Internat. Phonologie-Tagung / [Inst. für Sprachwiss. d. Univ. Innsbruck].
— Innsbruck: Inst. für Sprachwiss.
Bd. 1972 im Fink-Verl., München, Salzburg
NE: Internationale Phonologie-Tagung; Institut für Sprachwissenschaft ⟨Innsbruck⟩
4. Wien, 29. Juni — 2. Juli 1980. — 1981.
(Innsbrucker Beiträge zur Sprachwissenschaft; Bd. 36)
ISBN 3-85124-564-4

1981

INNSBRUCKER BEITRÄGE ZUR SPRACHWISSENSCHAFT

Herausgeber: Prof. Dr. Wolfgang Meid
Institut für Sprachwissenschaft der Universität Innsbruck
A-6020 INNSBRUCK, Innrain 52

Druck: Ernst Becvar, A-1150 Wien

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Paul Kiparsky

Remarks on the Metrical Structure of the Syllable[1]

Metrical representations have proved illuminating in the analysis of stress (Rischel 1964, 1972, Liberman and Prince 1977, Halle and Vergnaud 1979, Safir (ed.) 1979, Hayes 1980) and it is natural to inquire whether they are also appropriate for other prosodic phenomena. A prime candidate here is syllabicity, which shares with stress its essentially relational, syntagmatic character as well as the apparent lack of any invariant defining phonetic correlates. The purpose of this report is to discuss some of the advantages that may be gained from a metrical treatment of syllable structure.

Following Selkirk (1980) and others I shall assume that phonological theory provides a fixed hierarchy of prosodic levels, including at least those given in (1):

(1) phonological phrase
word
foot
syllable
segment

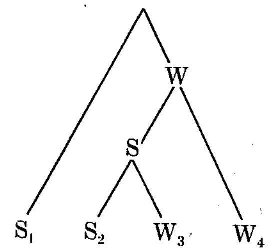
Thus, an element at each level is composed of one or more elements at the next lower level.

The first question is how structure at each level is to be represented. Here I shall be arguing for the position that each level is represented in a formally parallel fashion, by means of binary trees, each non-terminal node branching into *S(trong)* and *W(eak)*. This immediately forces me to take a position regarding the proper interpretation of such relational representations. What prominence relations do they induce among terminal nodes? Carlson (1978) and Hayes (1980) have noted that the original proposal by Liberman and Prince (1977) imposes an overly differentiated prominence pattern, and have adopted the weaker convention (2):

(2) The beat of a subtree labeled S is stronger than the beat of its sister subtree labeled W.

where the *beat* of a subtree is the terminal node connected to its root only via S nodes (the Designated Terminal Element of Liberman and Prince). Prominence relations will thus be established between two Strong terminal nodes (e. g. between nodes 1 and 2 in (3)) and between a Strong and a Weak terminal node (e. g. between 1 and 3 and between 1 and 4) but not between two Weak nodes (3 and 4).

[1] This work was supported in part by the National Institutes of Mental Health Grant No. 5 PO 1 MH 13390-13.

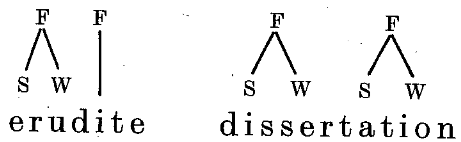


This convention will also prove appropriate at the level of syllable structure.

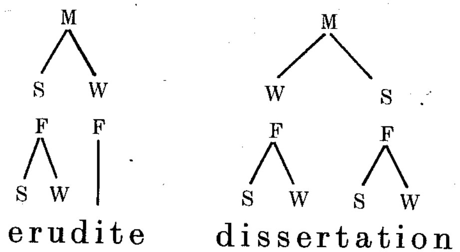
There are a number of reasons for assuming a hierarchy of prosodic levels such as (1).

(a) Stress in languages is assigned typically by distinct rules for each level. In English (grossly simplifying) syllables are organized into maximally binary feet labeled Strong-Weak[2], feet into right-branching word trees whose right branches are labeled Strong if branching and Weak if nonbranching, and finally word trees are organized according to the syntactic constituent structure into phrasal trees whose right branches are labeled Strong:

(4)



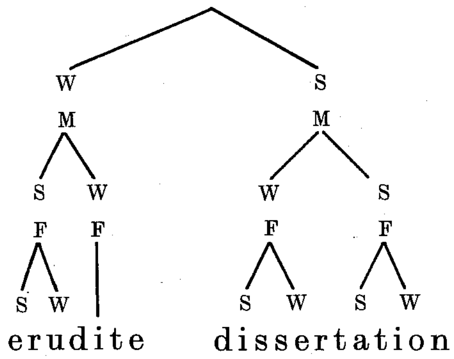
(foot level)



(word level)

[2] Where the right branch must be non-branching (i. e. a weak syllable). Here I follow Hayes (1980), who has made the surprising discovery that ternary feet in English words all result from later reorganization of the metrical structure by several distinct processes.

[3] This is not to exclude the possibility that two levels may coincide in some languages, e. g. the syllable and the foot (in "syllable-timed languages"), or the foot and the word.



(phrase level)

It is not always the case that rules are confined to only one level; we shall see examples below where rules apply within both syllables and feet. It does seem to be true that rules apply within domains delimited by these levels and not other kinds of domains.

(b) The basic quantitative opposition, which is branching vs. nonbranching, has to be defined at an appropriate level. Rules that form word trees look at whether *feet* are branching (*érudìte* vs. *èrudítion*), rules that form feet look at whether *syllables* are branching (*Cánada* vs. *Canásta*), but rules that form word trees do not care about branching *below* the foot level. E. g. the last foot of *erudite* is nonbranching for purposes of the word tree because it consists of one syllable, even though that syllable itself has a branching rhyme (i. e. is closed). This effect is reminiscent of syntactic principles such as subjacency (the "A-over-A constraint").

(c) The prosodic levels also constitute natural domains of *segmental* phonological rules, a commonplace except perhaps in regard to the foot level (Kiparsky 1979, Selkirk 1980, Prince 1980).

(d) By introducing a universal hierarchy of levels certain anomalous phonological features can be eliminated. To be *stressed* is to be the *beat of a foot* (Selkirk 1980, Prince 1980) and to be *syllabic* is to be the *beat of a syllable* (Kiparsky 1979, Kaye and Lowenstamm 1979). If metrical structure is extended below the segment level it may also be possible to eliminate the feature of length, representing e. g. long vowels as branching nuclei. (See Ingria 1980, Leben 1980 for related ideas). This meshes with the conclusion reached in recent autosegmental phonology that tones are not to be viewed as features but as "autosegments" on a separate tier of the phonological representation. The conjecture is thus not implausible that *there are no "suprasegmental" features.*

(e) As might be expected, poetic meter reveals the prosodic levels of language in a particualrly transparent way. For example, the metrical constraints governing the alternation of Strong and Weak positions are almost the same in Hopkins' so-called "Sprung Rhythm" as they are in standard English meters (Kiparsky 1977); what differs is principally the *level* in the metrical hierarchy to which the constraints apply.

With this general picture as background, let us consider how the basic facts of syllable structure could be accounted for. It appears that three sorts of constraints are involved. In English, if we exclude for the moment the "affix" of coronal obstruents (Fujimura 1979, see below), the post-nuclear consonant cluster, or "coda", is restricted as follows:

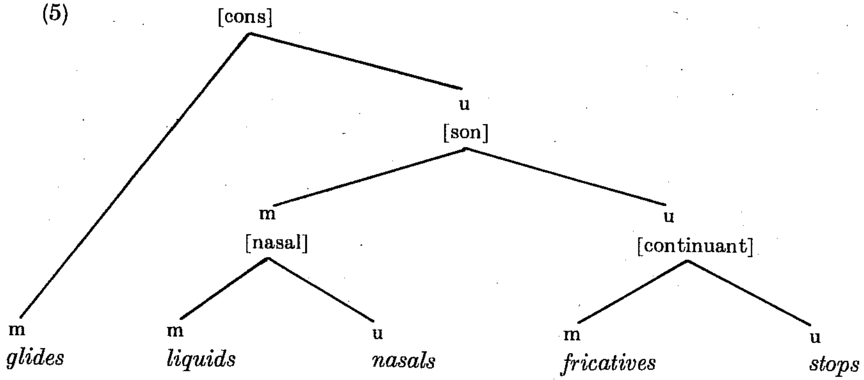
(i) *Length*: the coda can contain at most two segments. Hence *el, eyl, eym, eyp, elm, elp, emp* are possible rhymes of syllables, but **eylm, *eylp, *eymp, *elmp, *eylmp* are not possible.

(ii) *Sonority*: the relative prominence or sonority (Jespersen) must decrease from the nucleus towards the margins. The clusters in (i) are all compatible with this requirement, but e. g. **eml, *epl, *ely, *epm, *emy* are not.

(iii) *Idiosyncratic systematic gaps*: English happens not to allow e. g. voiced obstruents after nasals (**imb, *ing*) or after [u] (**ub, *ug, *uv*), even though such rhymes meet conditions (i) and (ii).

Of these three constraints, (iii) is wholly language-particular while (i) is a parameter whose particular value must be fixed for English. As for (ii), I assume that the sonority hierarchy can be formulated in universal phonetic terms, rather than ad hoc for each language. Fujimura's concept of *vowel affinity* (Fujimura 1979) is relevant here. Farmer (1979) has argued that the sonority hierarchy is derivable from the markedness principles and the universal hierarchy of major class features. As shown in (5), the degree of markedness of the major segment types corresponds exactly to the empirically postulated sonority hierarchy. Cf. also Basbøll 1977.

(5)



The syllable structure of all languages thus reflects an interplay of universal and language-particular constraints and tendencies (cf. Drachman 1977, Bell & Hooper 1978b). It is well-known that languages often have more or less limited types of violations of the sonority hierarchy, such as initial *st- sp-, sk-* in English or initial *rt-, mg-* in Russian or Polish. The same sequences might be impermissible as onsets in other languages; for example *rta*, monosyllabic in Russian, would be disyllabic in Serbocroatian, Czech, and Sanskrit. I have not found any phonetic data to show that there are any consistent physical correlates of this different syllabic valuation, and it would not be surprising to find that there are none. On the other hand, there is apparently always *phonological* motivation for syllabifications that violate the universal sonority hierarchy. Thus, whether the type *rta* is perceived as having one syllable or two depends upon whether the sonorant can bear the accent (as in Serbocroatian, Czech, or Sanskrit) or not (as in Russian and Polish). English *s-* is not considered a syllable because it cannot be stressed. Swedish *-ry* (e. g. in *berg* [bæry] 'mountain') is perceived as a coda because the syllabic valuation predicted by sonority would involve a stressed short syllable (*[bǽ-ri]), which is an impossible syllable structure in Swedish.

These considerations suggest that the phonological system is the primary determinant of language-particular aspects of syllable structure. More precisely, let us put forward the following two theses:

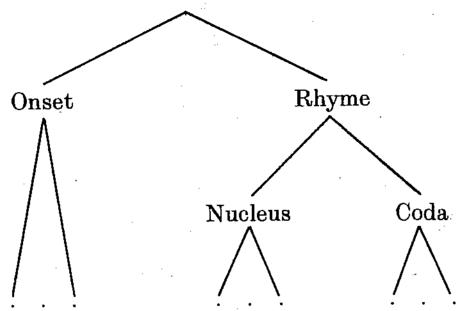
(5) Syllabification obeys the sonority hierarchy unless the phonological system of the language dictates otherwise.

(6) Syllabification cannot be determined purely from phonetic information.

If both (5) and (6) are true, it should follow that phonetically identical strings could be syllabified differently in languages with different phonological systems.

Kuryłowicz (1948), Pike and Pike (1948), and Hockett (1955) have argued on a variety of grounds that the syllable has an internal constituent structure of the form (7).

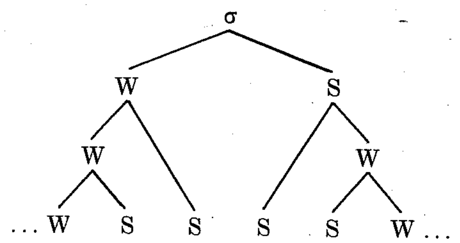
(7)



One development of this idea is to take the structure of (7) as a syllable template in universal phonetics. The labels "Onset", "Rhyme" etc. would thus be added to the set of elementary phonetic categories. This would be the position of Halle and Vergnaud (1979) and Selkirk (1980). Another approach is to take the universal syllable template as a relational structure represented by metrical trees of the familiar type. The terms "Onset", "Rhyme" etc. would then be defined derivatively in terms of the metrical structure. This relational approach is explored e. g. in Rischel (1964), McCarthy (1979), Kiparsky (1978, 1979), and Prince (1980).

Let us consider the following version of the relational approach. Syllabification rules assign binary branching metrical trees to strings of segments. The syllabification rules of a language consist of a universal core and language-particular specifications of certain general parameters as well as any idiosyncratic constraints on syllable structure. The universal template (8) which defines relative prominence on the segments according to (2) that matches the relative sonority of those segments according to the universal hierarchy (partially) represented in (5).

(8)



The prominence profile of the syllable, which decreases from the nucleus towards the margins, then follows directly from the interpretation assigned to subtrees of (8) by (2). Thus, "sonority" is simply the intrasyllabic counterpart of stress.

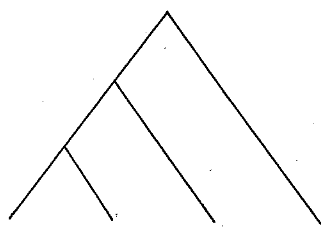
The length restriction mentioned above then is that the coda can branch only once. The idiosyncratic exemptions and restrictions will of course require special provisions in the grammar. It will be seen that the severity of co-occurrence restrictions increases with each node going down from the root.

Any theory must specify at what point in the derivation syllable structure is to be assigned. There are indications that it is done cyclically in English (Kiparsky 1979). In any case there is ample evidence from languages that syllable structure can be assigned early in the derivation, both because phonological rules can operate in syllabic environments (e. g. consonant gradation, cf. Prince 1980 for discussion) and because syllabic structure can depend upon phonological information not available at the phonetic level.

What are the reasons for assuming a relational representation of syllable structure?

(a) The relational representation brings out an *inherent* connection between the constituent structure of the syllable and its sonority profile. The ascending sonority in the onset and the descending sonority in the coda can *only* be represented by a left-branching and right-branching constituent structure, respectively. Suppose, contrary to fact, that the constituent structure of the rhyme were left-branching:

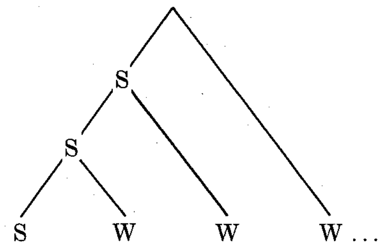
(9)



There would be no way to label the nodes in (9) so as to get falling sonority by the interpretive convention (2) or by any reasonable alternative to it. For example,

the labeling of (10) says that the relative sonority among all but the first segment is free, which is false:

(10)



Similarly, the representation brings out an *inherent* connection between the constituent structure and the unmarked status of CV syllables. $\widehat{\mathrm{W\ S}}$ is the first expansion of the node σ and therefore formally the simplest syllable structure. If we assume that the minimal structure is assigned, it follows from the form of the template (8) that VC syllables have an empty onset but CV syllables do not have an empty coda.

(b) The relational representation allows us to give a principled account of the opposition between heavy and light syllables. In so far as long vowels can be treated as geminates, the theory of syllable weight can be simply that syllables are heavy iff their rhymes (or nuclei) branch. Prosodic rules in general do not care whether there is an onset, or if there is one, what its internal structure is. Thus, we would have an adequate account of syllable weight if we had a principled way of not counting the onset as a branch. In a nonrelational theory there is no such principled way. It must simply be *stipulated* that the onset does not count. If the facts were different one could equally easily stipulate that rhymes or codas do not count. Therefore one does not have an explanation for what really is the case. In the proposed metrical representation, the corresponding stipulation is that a W normally does not count when it is a left branch:

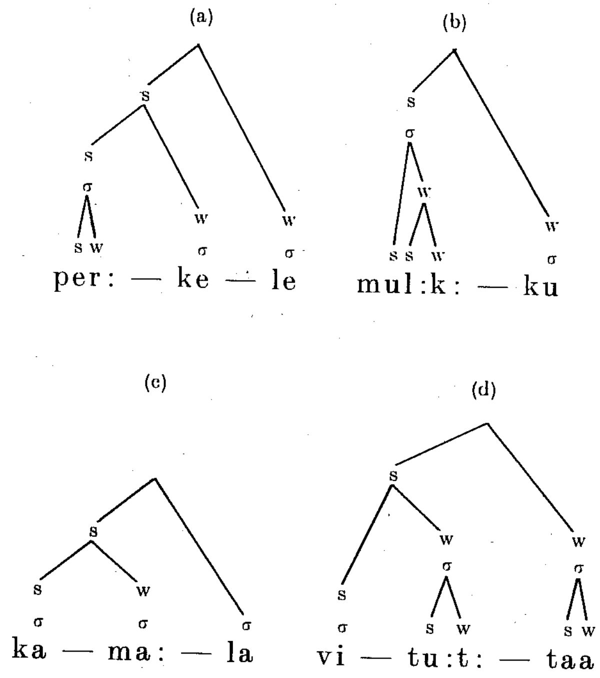
(11)

light (non-branching) heavy (branching)

In this form, the condition apparently recurs at higher levels of metrical structure, namely if the syllable node in (11) is replaced by a foot node. In languages with $\widehat{\mathrm{W\ S}}$ (right-dominant) feet the word tree is characteristically constructed without regard to branching. If this is the case, then extending relational metrical structure to the level of the syllable brings out a significant generalization[4].

(c) There are also language-particular rules that generalize across different levels of metrical structure, including the syllable. An example is the process of expressive lengthening in Finnish (Carlson 1978). What is lengthened is the W that follows the beat: that is, the coda if there is one (12a, b) and the whole next syllable otherwise, with onsets again not counting (12c, d):[5]

(12)



[4] At both levels, the omission of Weak left branches is apparently only a tendency, not an absolute necessity. There are languages in which the stress rules establish a quantitative distinction depending on whether the syllable has an onset or not (i. e. whether σ itself, not the rhyme, is branching). For Uradhi, an Australian language, Hale (1976) reports: "Uradhi stress is assigned by means of a rule which is apparently shared by all of the Northern Paman languages, namely $V \rightarrow \acute{V} \,/\, \#\,(V)\,C_1 \_\_\_$"

E. g. *utágak* 'dog', *mínhitji* 'bird'. That is, stress is assigned by a left-branching tree where left branches are labeled S if and only if they branch. At the foot level, $\widehat{W\,S}$ feet are quantitatively distinguished from monosyllabic feet in Cree (Halle and Vergnaud 1979) and Aklan (Hayes 1980). Arabic may also be a case (McCarthy 1979) but an independently motivated reanalysis due to Prince eliminates it (Hayes 1980: 391—182).

[5] Unfortunately it shows up best in curses and obscenities and I shall therefore omit glosses on the cited words.

Observe that in (12b) both segments in the cluster *lk* are lengthened, showing the right-branching constituent structure. If syllables have relational structure, this kind of process should be a natural one. In fact, the famous Estonian third quantity seems to be a result of this kind of lengthening process (Prince 1980).

(d) Distributional restrictions have an irreducibly relational character. For example: English *ink*, *imp* show that the first position in the coda has to admit nasals; *girl*, *file*, *owl* show that the second position has to admit *l*; yet **inl*, **iml* are not possible monosyllables. Why? Evidently because *l* is more sonorous than the nasals. That is, what can go in the second slot of the coda depends relationally on what is in the first. An approach that puts *absolute* restrictions on how the nodes of a structure such as (7) are filled has to treat such facts as arbitrary exceptions[6]. Another triplet that makes the same point is *asp*, *ask* (first position admits *s*), *harm*, *town* (second position admits nasals), vs. **asn*, **asm* (out because the sonority profile is rising).

There is a class of apparently exceptional codas which provide further evidence for the above view of syllable structure. With coronal obstruents, violations of each of the three types of restrictions are common:

(a) Violations of the length restriction: *wild*, *world*, *paint*, *exempt*, *lounge*, *texts*, (*thou*) *estrangedst*[7].

(b) Violations of the sonority profile: *apse*, *axe*, *bets*, *width*, *depth*, *paths*, *laughs*.

(c) Violations of the idiosyncratic restrictions: *bind*, *sand*, *mend* (voiced obstruent after a nasal); *wood*, *hood*, *could*, *should*, (voiced obstruent after [u]).

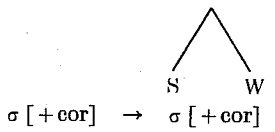
The long clusters of (a) could not be accomodated in the right-branching rhyme structure of (8) even if it were extended to the appropriate length, because the sonority profile would still be incompatible. The three exceptional aspects of dental obstruents in English syllable structure seen in (a), (b), and (c) follow at one stroke if the restrictions are left in force, and we adopt the idea of Fujimura and Lovins (1978, 1979)[8] that the syllable can contain an "affix", or what I shall here refer to as a set of *extrametrical segments*. These can be adjoined to the syllable by a segmental analog of the processes of "stray syllable adjunction" already well-known to the theory (Liberman and Prince 1977, Hayes 1980):

[6] Mohanan (1979) has attacked the relational view of syllable structure proposed here on the basis of a set of erroneous "universals". Thus he asserts that obstruents may never be syllabic and that sonority reversals may involve obstruents (e. g. *wœks*) but never glides (e. g. *wœkw*). Syllabic obstruents are notoriously found in many American Indian languages of the Pacific Northwest (Hoard 1978) and glides may idiosyncratically violate the sonority hierarchy in exactly the same way as obstruents: "Tamazight also exhibits many sequences of obstr + liq #, obstr + nasal #, and obstr + glide sequences, which according to the above strength hierarchy should not occur" (Saib 1978), e. g. *asy* 'take', *ašy* 'to be aware of'. A theory which predicts that these things cannot exist is *eo ipso* falsified.

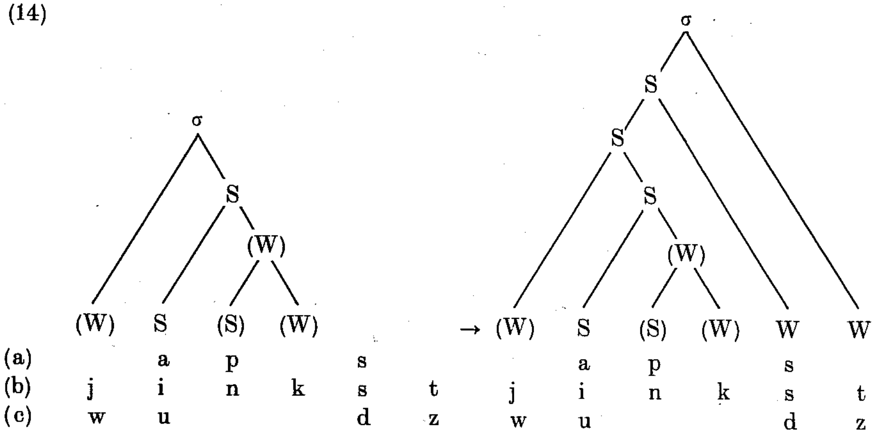
[7] The last example (which I owe to Jane Simpson) is probably a record for English (six consonants in the final cluster).

[8] Fujimura and Lovins motivate this treatment on phonetic grounds for a somewhat different class of obstruents. Halle (*voce*) has independently worked out a similar treatment for German based on Moulton (1956). See now Halle and Vergnaud (1980).

(13)



(14)



The sonority relations prohibit the *s* of *apse* from being accommodated in the core of the coda, but by virtue of (13) it can be accommodated as an extrametrical segment. The convention (2) that we adopted for interpreting the relative sonority assigned by metrical trees guarantees that there is no metrical prominence relation between the extrametrical W segments and the adjacent W inside the coda. Therefore the analysis correctly predicts the possibility of apparent violations of the sonority hierarchy.

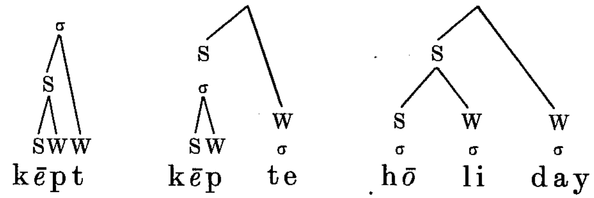
The adjunction rule (13) is the most direct way of accounting for the fact that the basic restriction to *two* postnuclear consonants is waived for coronal obstruents. (13) states no limit on the number of the extrametrical consonants. In practice the maximum is probably three (*texts*, *sixths*), though perhaps this is not an inherent phonological limit but follows from the limitations of English morphology.

At the same time, the rule eliminates the exceptionality of the coronal obstruents with respect to the idiosyncratic distributional restrictions such as the lack of voiced obstruents after nasals or [u]. These constraints hold within the coda and extrametrical segments are therefore not subject to them.

The several criteria we have found for internal syllable structure supply independent tests of the claim that certain segments are to be treated as extrametrical. Thus, by the same reasoning as above, the final consonant in words like *kēpt* would have had to be extrametrical in Middle English also (sonority would otherwise be violated). This structure allows us to derive the well-known shortening processes which took place at that period by a unitary metrical rule. Vowels were shortened before clusters, tautosyllabic or not, and before at least two syllables even if followed by only one consonant. As in Finnish expressive

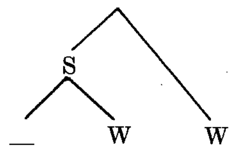
lengthening, right branches labeled W are metrically on a par, whether they are consonantal positions following the nucleus or unstressed syllables following the stressed syllable. The only difference is that while in Finnish W's are lengthened, here S's are shortened:

(15)



The underlined vowels in (15) are shortened by a rule applying in the environment

(16)



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Roger Lass

Explaining Sound Change: The Future of an Illusion*

1. Prologue. This is a programmatic and polemical paper: my concern is to establish a framework for debate rather than to follow arguments to their conclusions. (In any case the major arguments in §§ 2—3 have been spelled out in detail in Lass 1980, and some of those in § 4 at least adumbrated.)

As linguists we are public figures with pretensions to expertise, hence with the responsibilities of experts. That is, we are publicly accountable for the assertions we make (factual, explanatory, whatever). I assume that these ought ultimately to be judged by their warrantability: i. e. the degree of well-founded belief they can generate in a rational[1] hearer. In this light any proposition or conjunction of propositions p that purports to be an explanation of something can be reconstructed as an (alethically or epistemically) modalized statement N(p), P(p), $\sim$ P(p), etc., (where N = 'necessarily true' / 'confirmed beyond reasonable doubt', P = 'possibly true' / 'confirmed to some degree D < N', etc.). I further assume a hierarchy of excellence where $N > P > \sim P$.[2]

* This paper owes much to those mentioned in the acknowledgements to Lass (1980), since a good bit of the material in the earlier sections derives from that source. I am also indebted for (what I hope are) some clarifications due to oral presentations and discussions following the completion of the MS of that book: particularly to Jim Hurford, Steve Harlow, Jim Miller, and Suzanne Romaine. The final sections here draw heavily on discussions with Suzanne Romaine and suggestions by Geoff Sampson; the crazier ideas are basically my own. I am also grateful for suggestions and ideas at the Conference from Wolfgang Dressler, Henning Andersen, and Gaberell and Kiki Drachman. The notes below are rather more detailed responses to some issues that were raised in these discussions, as is the Epilogue.

[1] I am perhaps using a somewhat parochial definition of 'rational' here: one based on a neo-positivist and Western concept, devoid of any hint of cultural or historical relativity. Thus the fact that other sets of beliefs may be coherent and compelling of belief to others does not affect the rather puritanical point of view I adopt here (at least as a heuristic device). Thus it is irrelevant that some (e. g. the Zande or some Englishmen) find witchcraft a compelling framework for a world-view. I don't think that the arguments against the kind of monism I propose here (e. g. in Feyerabend 1975) destroy my basic point: at least not as a ground-clearing approach. The fact that any other system of beliefs may be compelling to its adherents as some form of deductivism/falsificationism may be to its adherents is simply irrelevant.

[2] This hierarchy could be condemned as unacceptably monistic and even 'scientistic'. Maybe it is; but if the point in question is purely *epistemic* excellence, then it does have a certain (if limited) power of compulsion compared to any 'non-standard' criterion of excellence. The question of appropriateness to the human sciences is another matter entirely: if historical linguistics is, say, parallel to human historiography rather than biological or cosmological historiography, then a positivist model is simply irrelevant — as is the question of explanation. I will argue below that explanation (in any sense) is irrelevant to historical linguistics: but not because it's a 'human science' like history. See the Epilogue for details.